Lecture 10 (Wulff & Langmuir)

Tuesday, 14 February 2023 10:30 AM

Surface Thermodynamics / Free energy

$$\gamma = \frac{1}{A}\left(U_{DFT} - TS_{excess} - \sum \mu_{metal} N^{metal}_{slab} \right) \quad ; \quad \mu_{metal} = g^{metal}_{bulk} \approx e^{DFT}_{bulk}$$

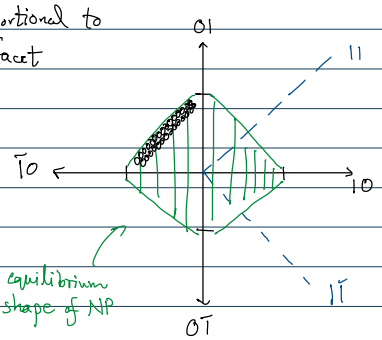
γ: Free energy change

$TS_{excess} \approx 0$ (first approx.)

Approach: Calculate γ's for different surfaces [(111), (100), (110), ...] → goal is to figure out which surface is present at equilibrium for catalyst nanoparticle.

Wulff Theorem (facet will be proportional to energy)

Distance from center of NP to a given facet is proportional to energy of that facet



$\gamma_{10} = 1$

$\gamma_{11} = 3/4$

$\gamma_{10} = \gamma_{01} = \gamma_{\bar{1}0} = \gamma_{0\bar{1}}$

$\gamma_{11} = \gamma \ldots$

(edges are the surfaces) — equilibrium shape of NP

Ref: Schall book Chp 2., 4.4

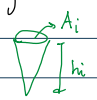
→ Doesn't say anything about how the particles are formed

"Proof":

$$G = \sum_i \gamma_i A_i \quad \text{(over facets)} \quad ; \text{ constant } V \rightarrow \text{const. bulk Energy}$$

Constraint: $V = \sum_i \frac{1}{3} h_i A_i$ (sum of a series of cones)

h_i: distance to center of NP



Lagrangian functional:

$$F = G - \lambda\left(\sum \tfrac{1}{3} h_i A_i - V\right)$$

$$\delta F = \delta G - \lambda \delta\left(\sum \tfrac{1}{3} h_i A_i\right)$$
$$= \delta\left[\gamma_i A_i - \lambda \tfrac{1}{3} \sum h_i A_i\right]$$
$$= \sum \delta\left[\gamma_i A_i - \tfrac{1}{3}\lambda h_i A_i\right] = 0$$

→ If the A_i can be varied independently → so each term is zero

$$\therefore \quad \gamma_i A_i = \frac{1}{3} h_i A_i$$

$$h_i = \frac{3\gamma_i}{\lambda} \quad \rightarrow \text{ from constraint: } \quad \lambda = \sum_i A_i \gamma_i$$

$h_i = \frac{3\gamma_i}{\lambda}$ → from constraint: $\lambda = \frac{\sum_i A_i \gamma_i}{V}$

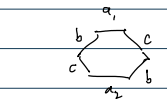
$$\boxed{h_i = \frac{3V}{\sum_j A_j \gamma_j}\gamma_i} \quad *$$

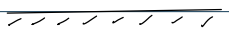
★ $h_i \propto \gamma_i$

*Not a full proof since we do not consider directions

Generalize Wulff to supports ⇒ adsorbates

1) Supports: start w/ a suspended NP





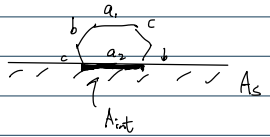
$$\Omega = 2\gamma_a\gamma_a + 2\gamma_b A_b + 2\gamma_c A_c + \underbrace{A_s\gamma_s}_{\text{constant}}$$

No interaction

$$\Omega - \gamma_s A_s = \underbrace{2\gamma_a A_a + 2\gamma_b A_b + 2\gamma_c A_c}_{\text{Standard Wulff construction}}$$

(⇒ we get same shape as unsupported case)

Interaction



$$\Omega = A_{a_1}\gamma_{a_1} + A_{a_2}\gamma_{a_2} + 2\gamma_b A_b + 2\gamma_c A_c + \gamma_{int} A_{int} + \gamma_c(A_s - A_{int})$$

$$\Omega - A_s\gamma_s = \gamma_{a_1} A_{a_1} + \gamma_{a_2} A_{a_2} + 2\gamma_b A_b + 2\gamma_c A_c + A_{int}(\gamma_{int} - \gamma_s)$$

$\gamma_{int} A_{int} = E_{int} - TS - \sum \mu_i N_i$ } XS quantities at interface

$\approx E_{int}^{DFT} - \mu_s N_s^{DFT} - \mu_m N_m^{DFT}$ } Total quantities in calculation

(support) $m = a_2$

Freestanding surfaces:

$$A_{int}\gamma_s = E_s^{DFT} - \mu_s N_s^{DFT}$$
$$A_{int}\gamma_m = E_m^{DFT} - \mu_m N_{m(a_2)}^{DFT}$$
} Freestanding surfaces

$$\Rightarrow A_{int}(\gamma_{int} - \gamma_{m,a_2} - \gamma_s) = E_{int}^{DFT} - E_s^{DFT} - E_m^{DFT} = \text{Adhesion Energy}$$

$$\Omega - A_s\gamma_s = \gamma_{a_1} A_{a_1} + 2\gamma_c A_c + 2\gamma_b A_b + A_{int}\left(\gamma_{m,a_2} + \frac{\Delta E_{adh}}{A_{int}}\right)$$

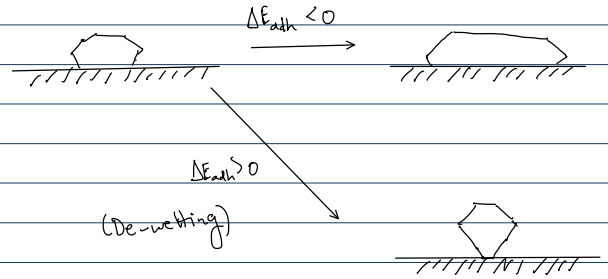
Wulff theorem

Effectively, replace γ with $\gamma + \frac{\Delta E_{adh}}{A_{int}}$

Effectively,

replace γ_{m,a_2} w/ $\gamma_{m,a_2} + \frac{\Delta E_{adh}}{A_{int}}$

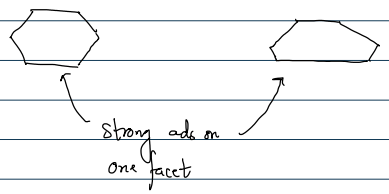
Eg:



(2) Adsorbates – "easy"; just replace $\gamma_{m,vacuum}$ w/ γ calculated from an ab-initio phase diagram

$\gamma(\Theta_i) \rightarrow \gamma(\mu_i) \rightarrow \gamma(T, P_i)$

Ex:



→ Re optimizing the shape of NP depending on the T, P

→ Assumption → (2) NP equilibrium

Why eq has high energy facets → Volume constraint!

(Need other facets or else will have higher bulk free energy → penalized)

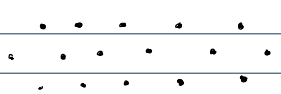
Isotherm Development

Pros: DFT calcs not necessary; better treatment of surface entropies

Cons: must parameterize surface energies and adsorption energies

(1) Langmuir isotherm → classic isotherm

Key assumptions: · no interaction b/w adsorbates
· uniform surface sites
· constant adsorption energies



+ Minimize Ω for given values of μ and T: $\Omega_A(\mu, T, N_s)$

+ Minimize Ω (Grand potential) for given values of μ and T : $\Omega(\mu, T, N_s)$ [A]

$\Omega = V - TS - \mu N = -k_B T \ln \Xi$ ← grand canonical partition function

$$\Xi = \sum_j e^{-\beta(E_j - \mu N_j)} = \sum_j e^{-\beta\left[\varepsilon(\sigma_1 + \sigma_1 \cdots \sigma_{N_s}) - \mu(\sigma_1 + \sigma_2 + \cdots \sigma_{N_s})\right]}$$

ε ← binding energy of 1 molecule

N_s → total sites

$\sigma_i = \begin{cases} 0 \\ 1 \end{cases}$ unoccupied / occupied So, $\sum_i^{N_s} \sigma_i = N$

$$\therefore \Xi = \sum_{\sigma_1=0}^{1} e^{-(\varepsilon-\mu)\sigma_1} \sum_{\sigma_2=0}^{1} e^{-\beta(\varepsilon-\mu)\sigma_2} \cdots \sum_{\sigma_{N_s}=0}^{1} e^{-\beta(\varepsilon-\mu)\sigma_{N_s}}$$

$$= \prod_{i=1}^{N_s} \sum_{\sigma=0}^{1} e^{-\beta(\varepsilon-\mu)\sigma}$$

$$= \prod_{i=1}^{N_s} \left(1 + e^{-\beta(\varepsilon-\mu)}\right)$$

$$\boxed{\Xi = \left[1 + e^{-\beta(\varepsilon-\mu)}\right]^{N_s}}$$

$$\therefore \Omega = -k_B T N_s \ln\left(1 + e^{-\beta(\varepsilon-\mu)}\right)$$

$$\langle N \rangle = -\frac{\partial \Omega}{\partial \mu} = \frac{e^{-\beta(\varepsilon-\mu)}}{1 + e^{-\beta(\varepsilon-\mu)}} \cdot N_s$$

$$\therefore \boxed{\frac{\langle N \rangle}{N_s} = \Theta = \frac{e^{-\beta(\varepsilon-\mu)}}{1 + e^{-\beta(\varepsilon-\mu)}}}$$

If surface is in equil. w/ ideal gas:

$$\mu = \mu_{gas} = \mathring{\mu}_{gas} + k_B T \ln\left(\frac{P}{P^\circ}\right)$$

$$\boxed{\Theta = \frac{\frac{P}{P^\circ} e^{-\beta(\varepsilon-\mu^\circ)}}{1 + \frac{P}{P^\circ} e^{-\beta(\varepsilon-\mu^\circ)}}}$$

Note: This assumes that S is only configurational entropy

ε can be replaced w/ surface free energy w/o losing generality

$$\varepsilon \rightarrow \varepsilon + ZPE + \text{thermal vibrations} - T S^\circ_{loc}$$

↑ Not configurational

$$\simeq \mathring{\mu}_{surf}$$

$$\therefore \theta = \frac{\frac{\rho}{\rho^\circ} e^{-\beta \Delta\mu^\circ}}{1 + \frac{\rho}{\rho^\circ} e^{-\beta \Delta\mu^\circ}}$$